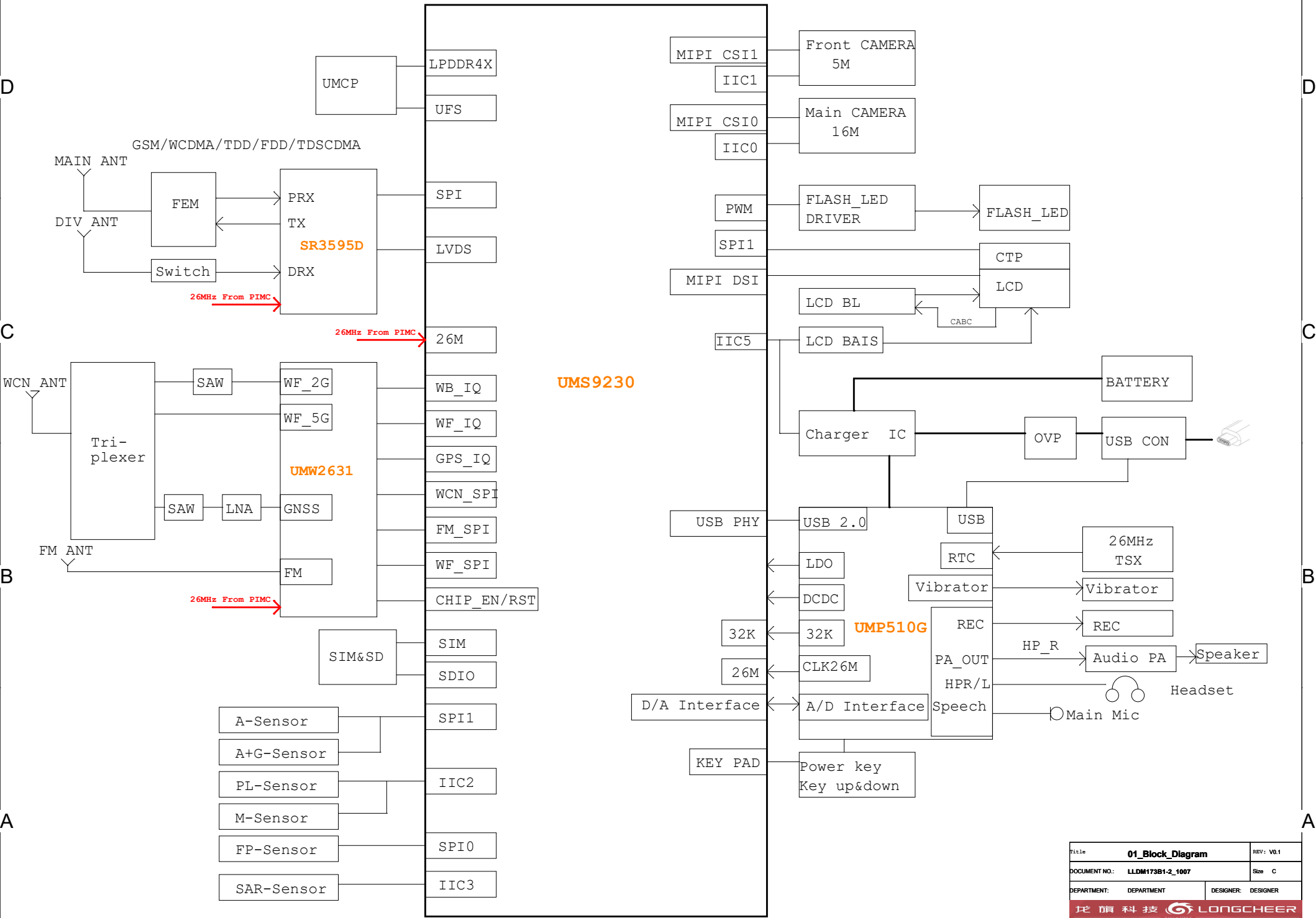


BLOCK DIAGRAM



REV	Description
REV0.1	Initial release

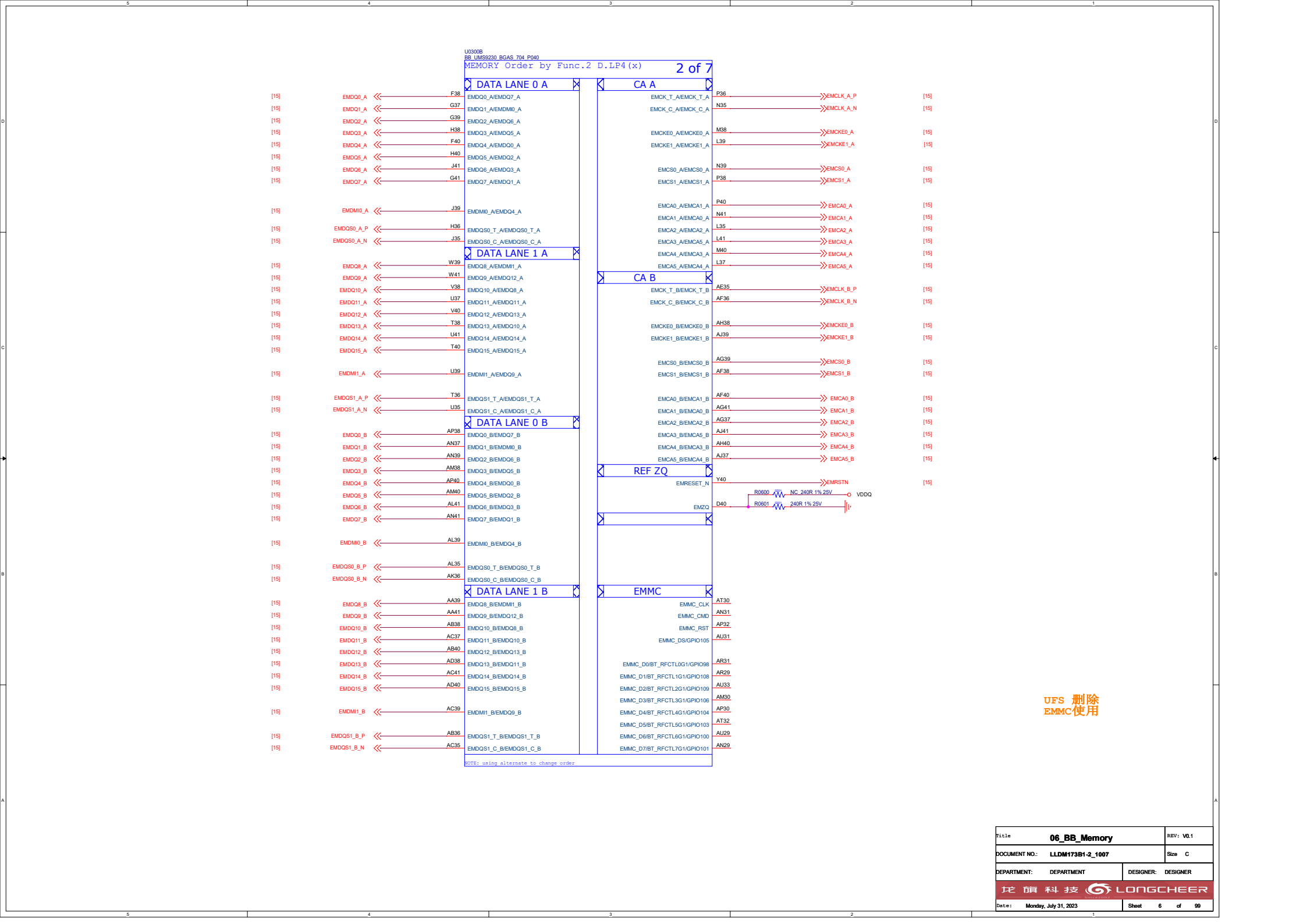
Title	02_BB_change list	REV: V0.1
DOCUMENT NO.:	LLDM173B1-2_1007	Size B
DEPARTMENT:	DEPARTMENT	DESIGNER: DESIGNER
龙 旗 科 技  LONGCHEER		
Date:	Monday, July 31, 2023	Sheet 2 of 99

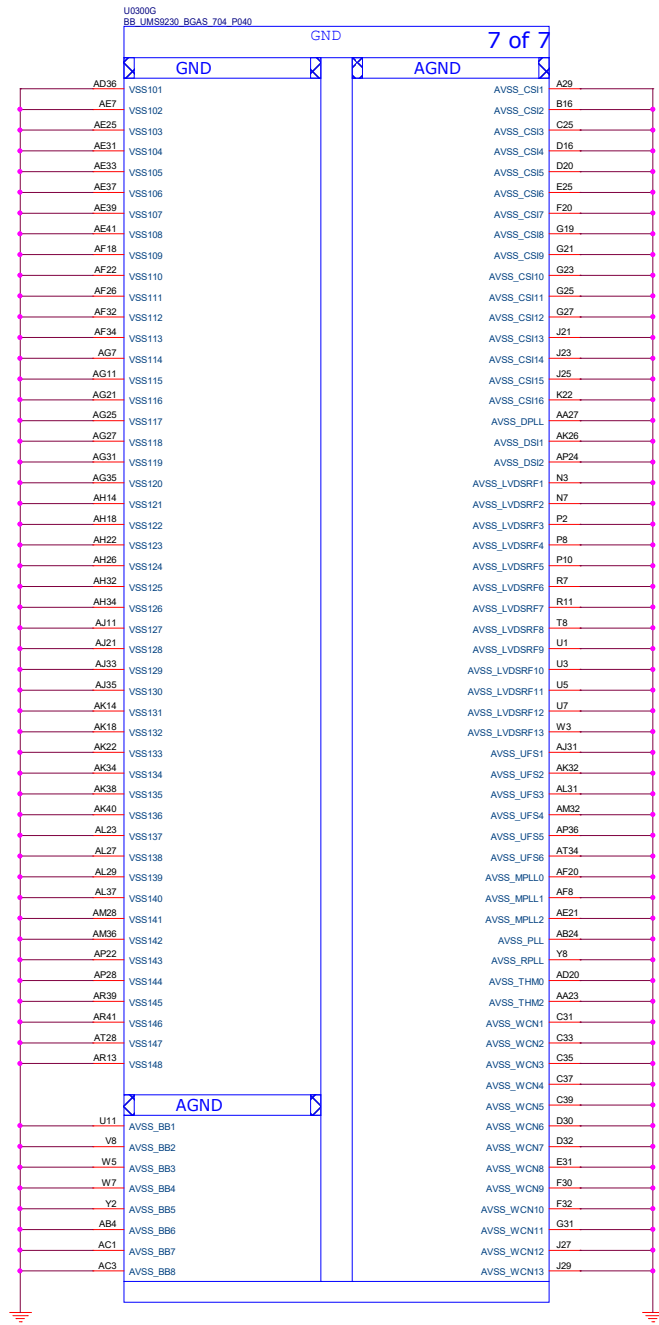
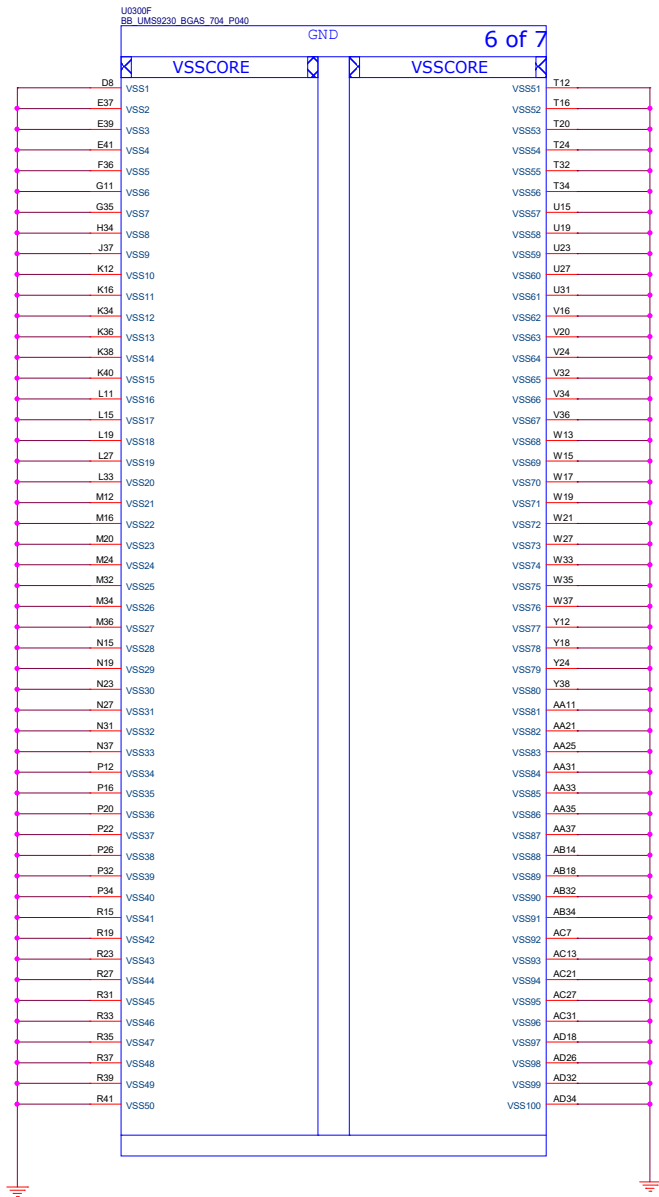


Ball name	Signal name in board code	Value	Description	Default value
UTTXD	USB_DLOAD_EN	0	USB download time out	1(=1 s)
		1	UNST	
AUD_DAD1	ARM_BOOT_MD0	00	Reserved	1(=1 s)
AUD_DAD0	ARM_BOOT_MD1	10	SD boot	(MBS-MD1-USB-MD0)
		11	MMC & SD	
ADI_SCLK	ARM_BOOT_MD2	0	Reserved	1(=1 s)
ADI_D	ARM_BOOT_MD3	0	Reserved	1(=1 s)
CHD_TYPE	ARM_BOOT_MD4	0	Reserved	1(=1 s)
		1	USB	

Front Camera

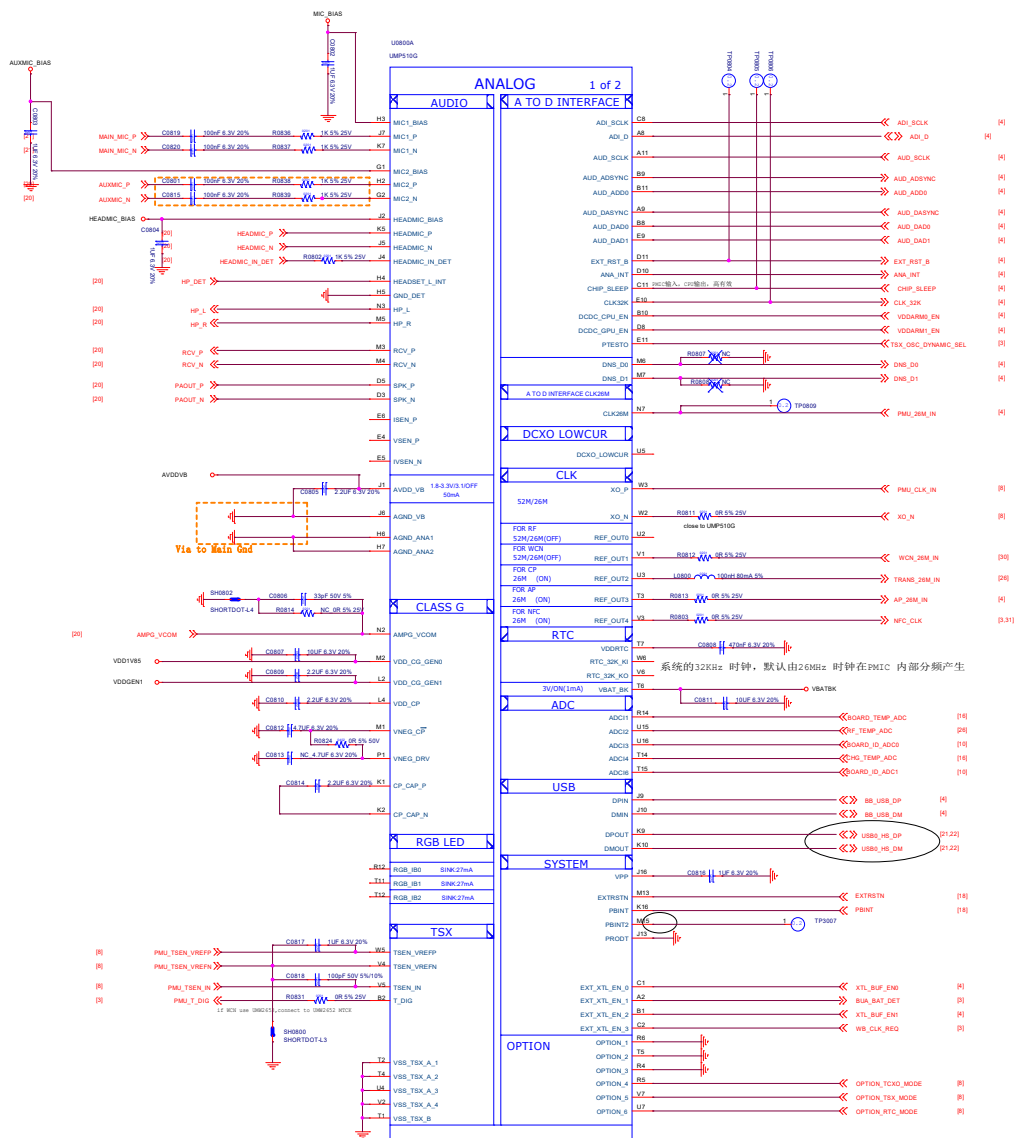
Main CAMERA



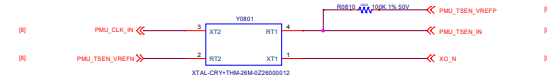


Title		07_BB_GND		REV: V0.1	
DOCUMENT NO.:		LLDM173B1-2_1007		Size C	
DEPARTMENT:		DEPARTMENT		DESIGNER: DESIGNER	
龙喙科技  LONGCHEER					
Date:		Monday, July 31, 2023		Sheet 7 of 99	

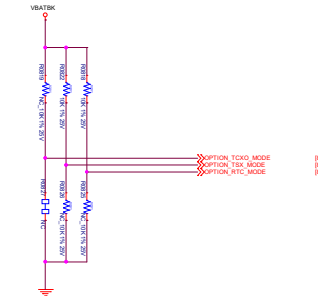
PMU_ANALOG



TSX (PMU) Only



Board ID

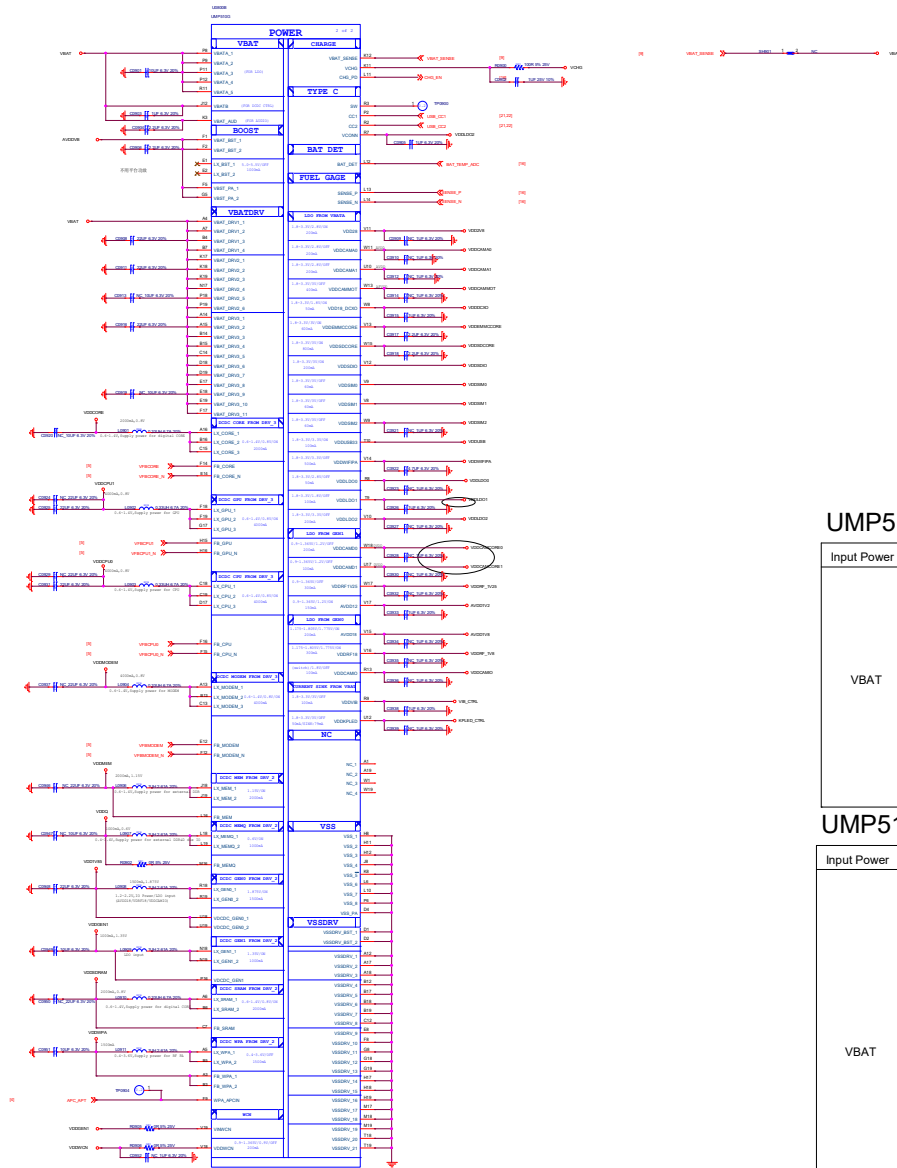


Item	Scenario	Option1	Option2
0	Qogir L6+UMP510G	0	0
1	Qogir L6+UMP510G+DCDC	0	1
2			
3			

Scenario	Option3(26M/32K)	Option4(TCXO_MODE)	Option5(TSX_MODE)	Option6(RTC_MODE)
26M TCXO+32K Crystal	0	1	0	0
26M TSX+32K Crystal	0	0	1	0
26M TSX+32K-less from PMIC internal	0	0	1	1
26M DCXO+32K-less from PMIC internal	0	1	1	1
26M DCXO+32K Crystal	0	1	1	0

Option1	Option2	Description
0	0	The DCDCGPU controlled by UMP510G ADI interface signal DCDC_GPU_EN. The typical application of DCDCGPU is Unisoc BB chip CPU power supply.
0	1	The DCDCGPU default on and uncontrollable by UMP510G ADI interface signal DCDC_GPU_EN
1	0	The DCDCMEMQ controlled by UMP510G ADI interface signal DCDC_GPU_EN. The typical application of DCDCMEMQ is Unisoc BB chip dual rail SRAM power supply.
1	1	

shipping mode



UMS9230	T606	不需要外部DCDC供电
UMS9230H	T612	VDD_CPU1使用PMIC DCDC_GPU供电
UMS9230T	T616	需要外部DCDC供电

Power name	Output Voltage(V)	Output Current	Default Voltage(V)	Default status	Feedback Sense	Usage
DCDCCORE	0.6-1.4	2000mA	0.8	ON	Remote	Supply power for digital CORE
DCDCGPU	0.6-1.4	4000mA	0.8	ON	Remote	Supply power for GPU
DCDCCPU	0.6-1.4	4000mA	0.8	ON	Remote	Supply power for CPU
DCDCMODEM	0.6-1.4	4000mA	0.8	ON	Remote	Supply power for MODEM
DCDCMEM	0.6-1.4	2000mA	1.15	ON	Local	Supply power for external DDR
DCDCMEMQ	0.4-1.4	1000mA	0.6	ON	Local	Supply power for external DDR & D die IO
DCDCGEN0	1.2-2.2	1500mA	1.875	ON	Local	IO Power/LDO input (AVDD18/VDDIF18/VDDCAM0)
DCDCGEN1	0.6-1.85	1000mA	1.35	ON	Local	LDO input (VDDCAM0/VDDCAM1/VDDRF1V25/AVDD12)
DCDCGRAM	0.6-1.4	2000mA	0.8	ON	Local	Supply power for digital CORE
DCDCWPA	0.4-3.6	1500mA	NA	OFF	Local	Supply power for RF PA
VBST_PA	5.0-6.0	1000mA	NA	OFF	Local	Audio PA Power

Power	Power Name	Output Voltage (V)	Output Current (mA)	Trim w/o (mV)	Default Voltage(V)	Default status	Usage
VBAT	VDDCAM0	1.8-3.3	1-2000	80.2	2.8	OFF	Camera analog power supply
	VDDCAM1	1.8-3.3	200	80.2	2.8	OFF	Camera analog power supply
	VDDCAMMOT	1.8-3.3	400	80.4	3.0	OFF	Camera Audio focus motor power supply
	VDDSDCORE	1.8-3.3	800	80.8	3.0	ON	SD Card power supply
	VDDSDIO	1.8-3.3	60	80.1	3.0	OFF	SD Card power supply
	VDDSIM1	1.8-3.3	60	80.1	3.0	OFF	SD Card power supply
	VDDSIM2	1.8-3.3	60	80.1	3.0	OFF	SD Card power supply
	VDDIOB	1.8-3.3	200	80.2	2.8	ON	IO LDO analog power supply
	VDDMEMCORE	1.8-3.3	600	80.6	3.0	ON	RAM core power supply
	VDDLDO0	1.8-3.3	50	80.1	2.8	OFF	GPIO LNA, RF switch
	VDDLDO1	1.8-3.3	100	80.1	1.8	OFF	Normally voltage mode for vibrator
	VDDLDO2	1.8-3.3	200	80.2	3.3	OFF	Normally voltage mode for vibrator
	VDDSDIO	1.8-3.3	200	80.2	3.0	ON	SDIO power supply
	VDDVIB	2.4-3.3	1	80.1	3.0	ON	Supply for RTC demand
	VDDVIB	1.8-3.3	100	80.1	3.0	OFF	Vibrator mode for vibrator
	VDDKPLED	1.8-3.3	50	80.1	3.0	OFF	Voltage mode for KPLED
	VDDWPA	1.8-3.3	500	80.0	3.3	OFF	WPA Power Amplifier (PA) supply
	VDDH_DCKO	1.8-3.3	50	80.1	1.8	ON	DCKO power supply
	VDDUSB33	1.8-3.3	100	80.1	3.3	ON	USB 3.0 (internal) power supply

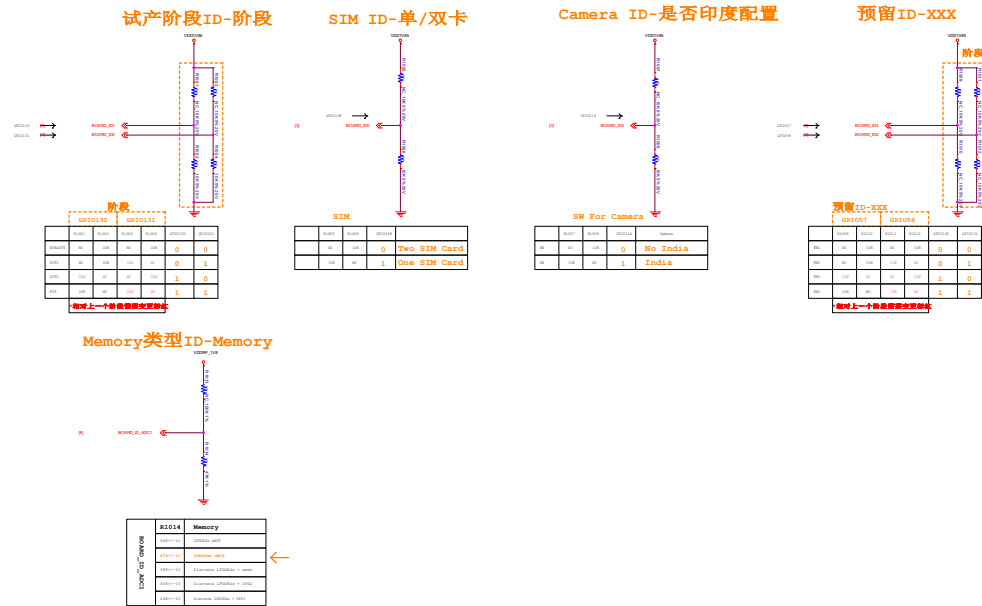
UMP510G5 DCDC output

Input Power	Power Name	Output Voltage (V)	Output Current (mA)	Default Voltage (V)	Default
VBAT	DCDC CORE	0.3-1.3	2000	0.8	ON
	DCDC GPU	0.3-1.3	4000	0.8	ON
	DCDC CPU	0.3-1.3	4000	0.8	ON
	DCDC MODEM	0.3-1.3	4000	0.8	ON
	DCDC MEM	0.25-1.25	2000	1.15	ON
	DCDC MEMQ	0.25-1.25	1000	0.6	ON
	DCDC GEN0	1.8-2.5	1500	1.85	ON
	DCDC GEN1	0.6-1.5	1000	1.3	ON
	DCDC SRAM	0.3-1.3	2000	0.8	ON
	DCDC WPA	0.6-3.3	1500	/	OFF

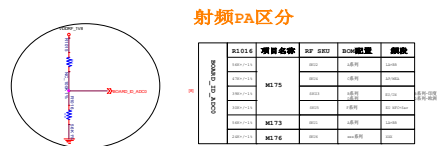
UMP510G5 LDO output

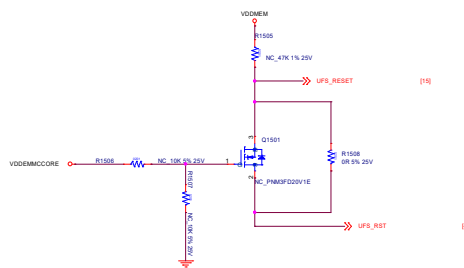
Input Power	Power Name	Output Voltage (V)	Output Current (mA)	Default Voltage (V)	Default
VBAT	VDD28	1.8-3.3	200	2.8	ON
	VDDCAMA0	1.8-3.3	200	2.8	OFF
	VDDCAMA1	1.8-3.3	200	2.8	OFF
	VDDCAMMOT	1.8-3.3	400	3	OFF
	VDD18_DCKO	1.8-3.3	50	1.8	ON
	VDDMEMCORE	1.8-3.3	600	3	ON
	VDDSDCORE	1.8-3.3	800	3	ON
	VDDSDIO	1.8-3.3	200	3	ON
	VDDSIM0	1.8-3.3	60	3	OFF
	VDDSIM1	1.8-3.3	60	3	OFF
	VDDSIM2	1.8-3.3	60	3	OFF
	VDDUSB33	1.8-3.3	100	3.3	ON
	VDDWIFIPA	1.8-3.3	500	3.3	OFF
	VDDLDO0	1.8-3.3	50	2.8	OFF
	VDDLDO1	1.8-3.3	100	3.3	OFF
	VDDLDO2	1.8-3.3	200	3.3	OFF
GEN1	VDDCAMD0	1.0-1.25	400	1.1	OFF
	VDDCAMD1	1.0-1.25	200	1.1	OFF
	VDDRF1V25	1.0-1.25	300	1.25	OFF
	AVDD12	1.0-1.25	150	1.2	ON
GEN0	AVDD18	1.2-1.8	200	1.8	ON
	VDDRF18	1.2-1.8	300	1.8	ON
	VDDCAMIO	1.2-1.8	100	1.8	OFF
GEN1	WCN	1.0-1.6	200	1	OFF

BB 部分



RF 部分





PMIC SW	PMIC VDD05MMIOCORE
UMPS100	3.0V/600mA

Notes: If DVS ICC is greater than 600mA, please use external LDO or DCDC.

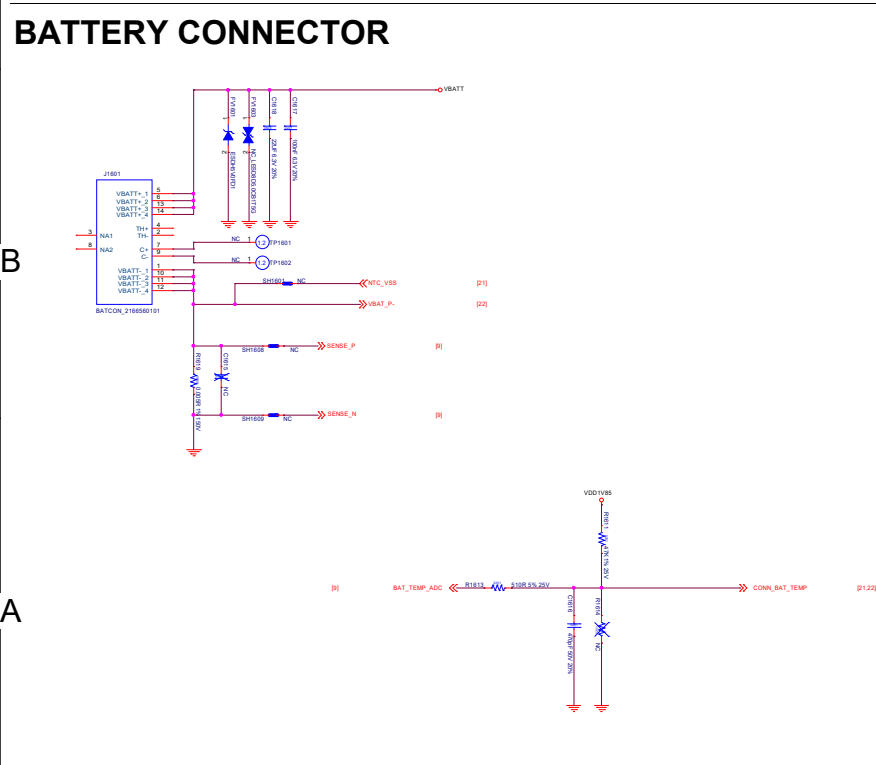
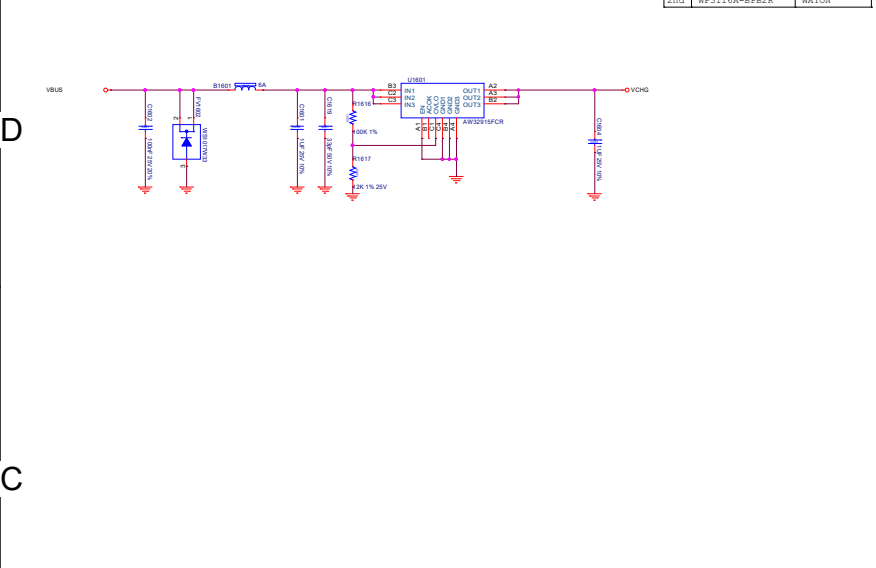
Title	15_uMCP_LPDDR4X	REV: V0.1
DOCUMENT NO.:	Design Name = LLEMY7301_2_1007	Size D
DEPARTMENT:	DEPARTMENT	DESIGNER: DESIGNER
		
Date:	Monday, July 31, 2023	Sheet 15 of 99

OVP

5

4

		U1601		Manufacture	1	P/N	Manufacture
1st		AW32915PCR		Awinic			
2nd		WP3116A-BPB2R		WATON			



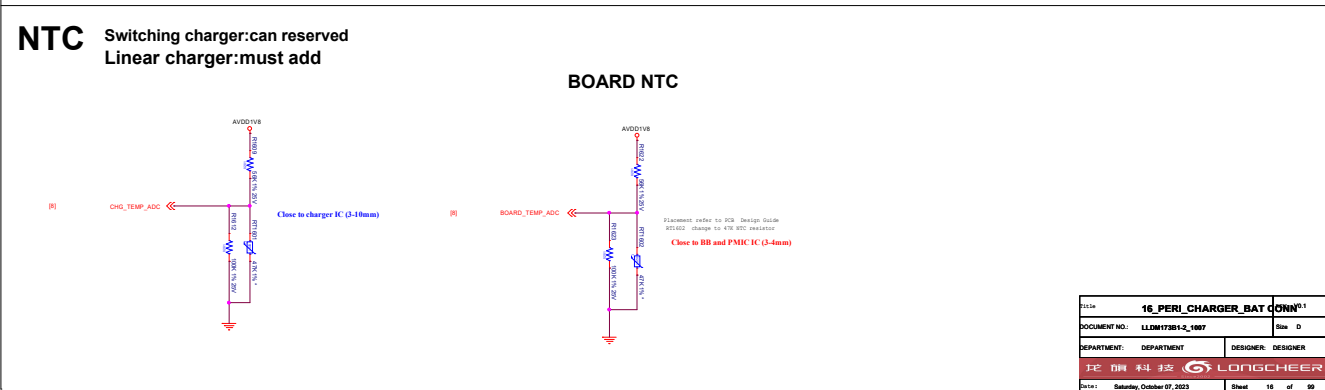
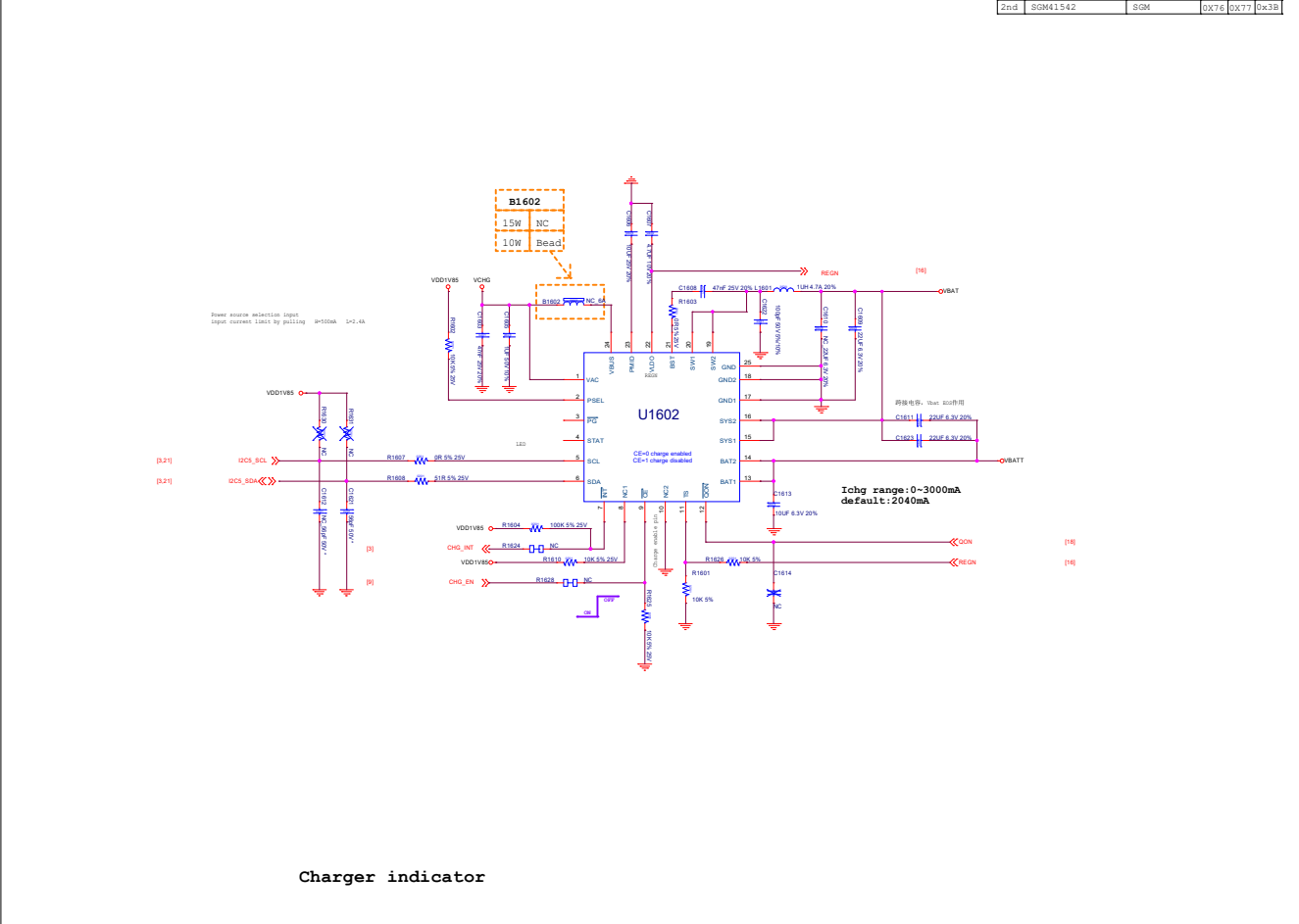
Switching Charger

3

2

1

		U1602		Manufacture	1	P/N	Manufacture	W	R	7bit
1st		UPK6920		Unisemi		0XD4	0XD5	0x6A		
2nd		SGM41542		SGM		0X76	0X77	0x3B		



REAR CAMERA: 13M

D

For 50M

C

FRONT CAMERA: 5M

B

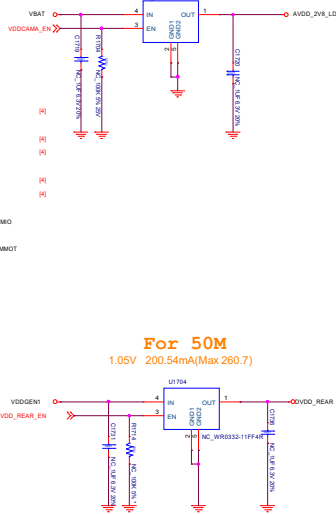
A

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	AVDD	DVDD	IOVDD	AFVDD
Typ	2.8V 33mA	1.1V 0.1mA	1.8V 1.7V	TRD
Min	2.7V 4mA	1.0V 0.05mA	1.7V 1.6V	TRD
Max	2.9V 4mA	1.2V 0.05mA	2.0V 1.8V	TRD

U1701:reserve for high PSRR VDDCAMA



For 50M
1.05V 200.54mA(Max 260.7)

	AVDD	DVDD	IOVDD	AFVDD
Typ	2.8V 33mA	1.1V 0.1mA	1.8V 1.7V	TRD
Min	2.7V 4mA	1.0V 0.05mA	1.7V 1.6V	TRD
Max	2.9V 4mA	1.2V 0.05mA	2.0V 1.8V	TRD

Macro Camera 2M

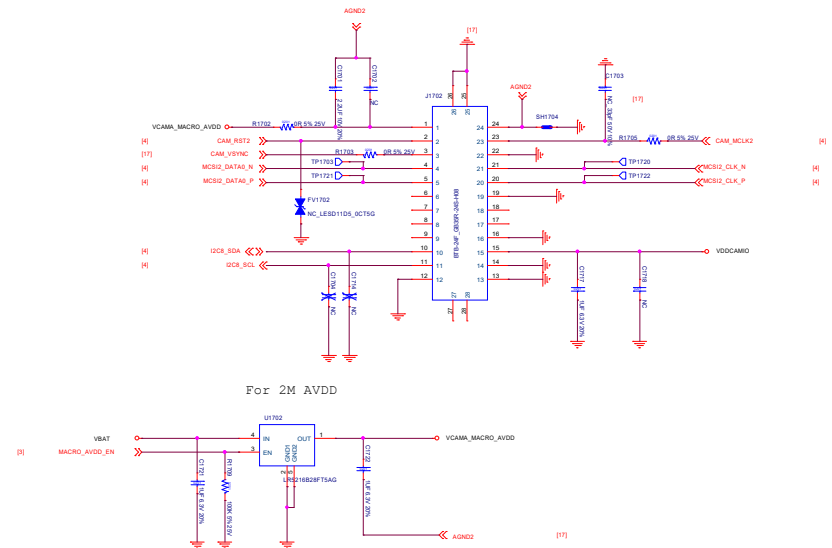
3

FLASH LED

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	AVDD	DVDD	IOVDD	AFVDD
Typ	2.8V 33mA	1.1V 0.1mA	1.8V 1.7V	TRD
Min	2.7V 4mA	1.0V 0.05mA	1.7V 1.6V	TRD
Max	2.9V 4mA	1.2V 0.05mA	2.0V 1.8V	TRD



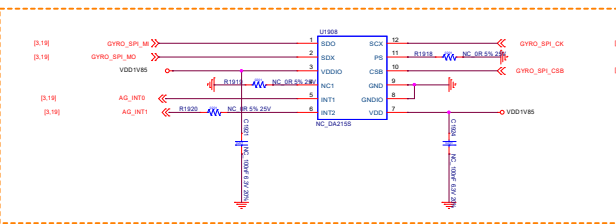
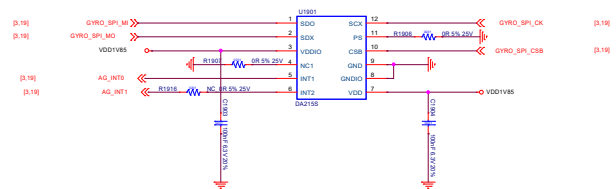
1_micron+00019_ledc2
1_flash+00019_ledc2
1_micron+00019_ledc2 + 2
1_flash+00019_ledc2 + 2
1_micron+00019_ledc2 + 2
1_flash+00019_ledc2 + 2

Connect to main ground plane by via directly

17_CAMERA		Rev: V0.1
DOCUMENT NO: LL0017381-2_1987		Rev: 0
DEPARTMENT: DEPARTMENT	DESIGNER: DESIGNER	
Date: Wednesday, August 01, 2023		Sheet: 17 of 99

A-Sensor⁵

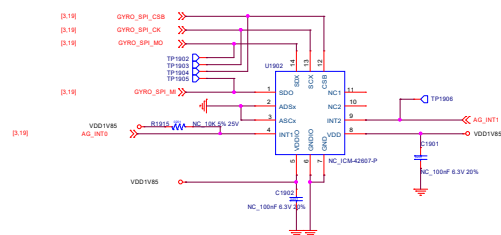
	U1901	4
1st	IMMU553	SoC/IC
2nd	STR5321	Sensortek



先摆放2个区域，后期根据位置环境选择一个，先做兼容设计

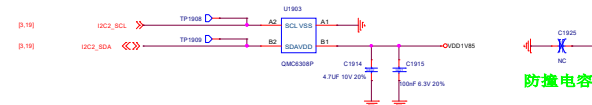
A+G-Sensor³

	U1902	Manufacture	1	P/N	Manufacture
1st	ICM-42607-P	TOR			
2nd	BM1320	ROHM			

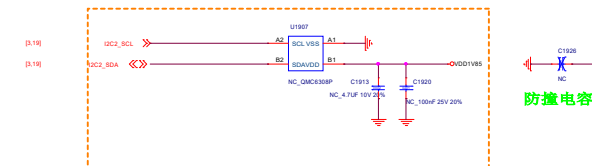


P+L Sensor

	U1903	1	I2C
1st	MM5603N3L	MEMSIC	0x60 0x61 7bit
2nd	QMC6308P	QST	0x58 0x59 0x2C



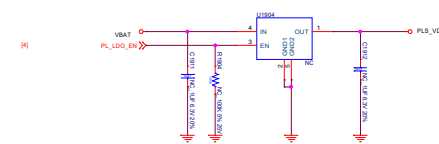
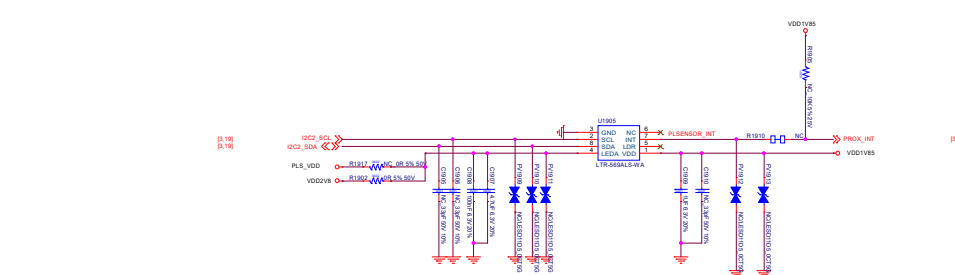
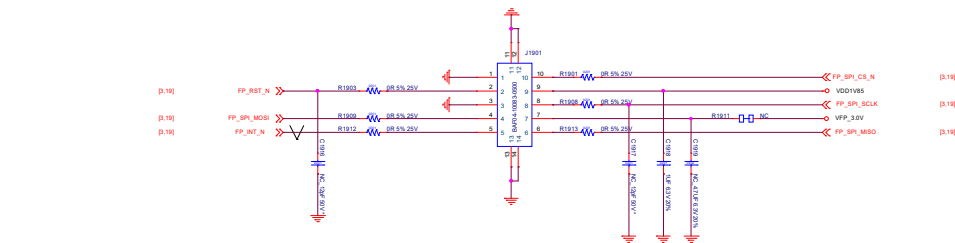
防撞电容



防撞电容

先摆放2个区域，后期根据位置环境选择一个，先做兼容设计

Finger Print



2114	19_SENSOR&VIBR	REV: V0.1
DOCUMENT NO:	LL0N17361-4_1907	Rev: 0
DEPARTMENT:	DEPARTMENT	DESIGNER: DESIGNER
龙 鼎 科 技 LONGCHEER		
DATE:	Saturday, October 07, 2023	Sheet 19 of 99

Audio PA⁵ For Speaker

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Manufacturer	FS1815N	Manufacturer	FS1815N
Part	FS1815N	Part	FS1815N
Rev	0A	Rev	0A

AD Pin	ADS2	ADS	I2C Address
/	0	0	0x34
/	0	1	0x35
/	1	0	0x36
/	1	1	0x37

Layout 器件T或L型，
大面积铺铜

Receiver

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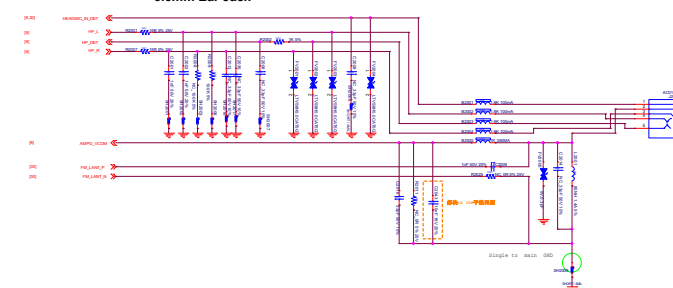
2

1

3.5mm Headphone Jack



3.5mm Ear Jack



Doc No.	2B_AUDIO	Rev	1.0
Doc Name	2B_AUDIO_1.0	Rev	1.0
Doc Type	PCB Layout	Doc Type	PCB Layout
Doc Date	2023-03-15	Doc Date	2023-03-15
Doc Author	Longcheer	Doc Author	Longcheer

A



A

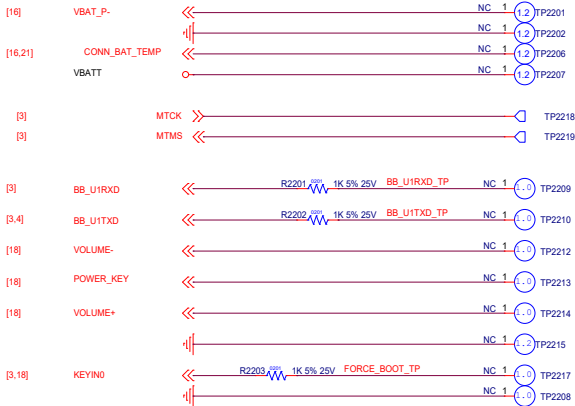


Title		21_SUB_LCM IF_LCMBIAS_VBL V0.1	
DOCUMENT NO.:		LLDM173B1-2_1007	Size D
DEPARTMENT:		DEPARTMENT	DESIGNER: DESIGNER
 LONGCHEER			
Date:		Wednesday, August 02, 2023	Sheet 21 of 30

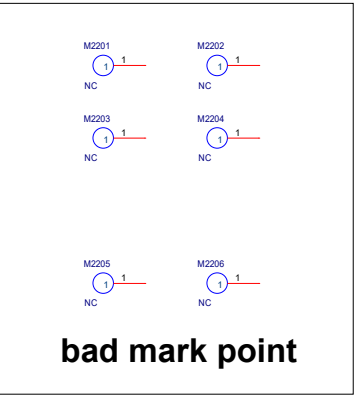
USB TEST POINT



BAT



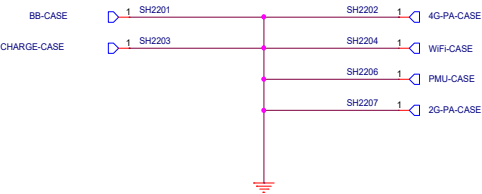
Marks point



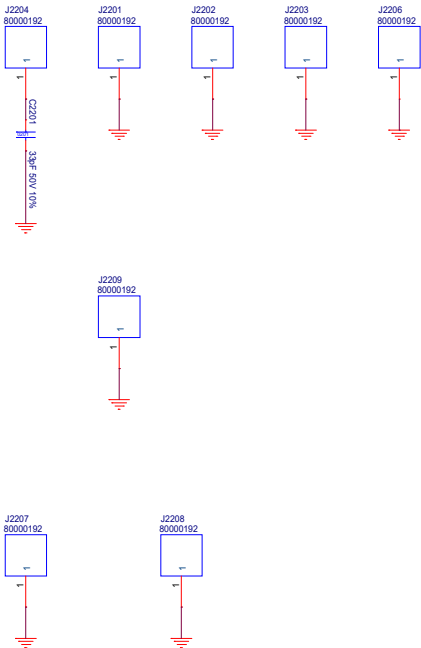
Shielding frame

TOP

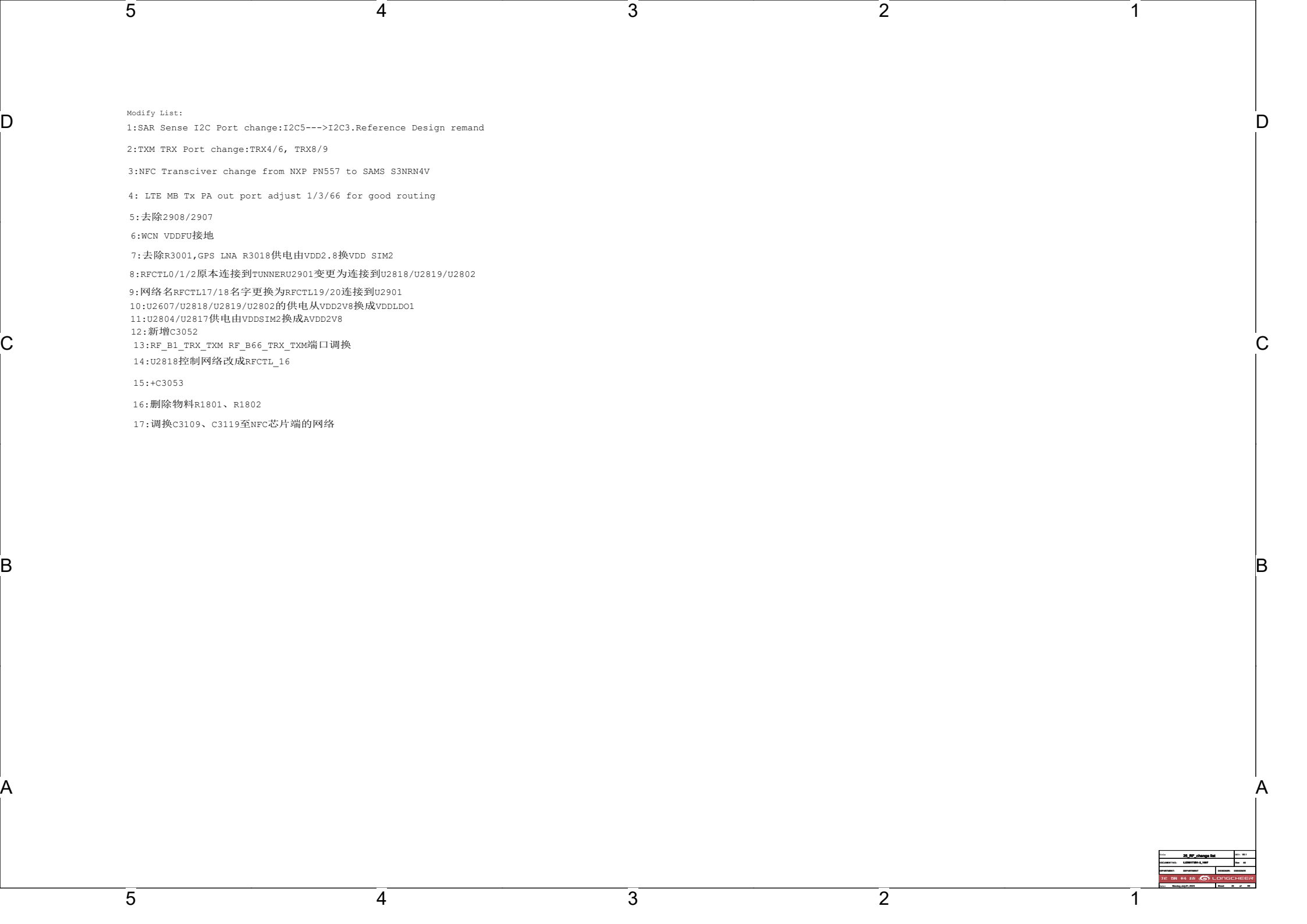
BOTTOM



接地弹片



Q2201&Q2202		
	Manufacture 1 P/N	Manufacture
1st	CJBA3134K	JSCJ
2nd	LSI1012BN3T5G	LRC
3rd	PNM3FD20V1EM	Prisemi



Modify List:

1:SAR Sense I2C Port change:I2C5--->I2C3.Reference Design remand

2:TXM TRX Port change:TRX4/6, TRX8/9

3:NFC Transciver change from NXP PN557 to SAMS S3NRN4V

4: LTE MB Tx PA out port adjust 1/3/66 for good routing

5:去除2908/2907

6:WCN VDDFU接地

7:去除R3001,GPS LNA R3018供电由VDD2.8换VDD SIM2

8:RFCTL0/1/2原本连接到TUNNERU2901变更为连接到U2818/U2819/U2802

9:网络名RFCTL17/18名字更换为RFCTL19/20连接到U2901

10:U2607/U2818/U2819/U2802的供电从VDD2V8换成VDDLDO1

11:U2804/U2817供电由VDDSIM2换成AVDD2V8

12:新增C3052

13:RF_B1_TRX_TXM RF_B66_TRX_TXM端口调换

14:U2818控制网络改成RFCTL_16

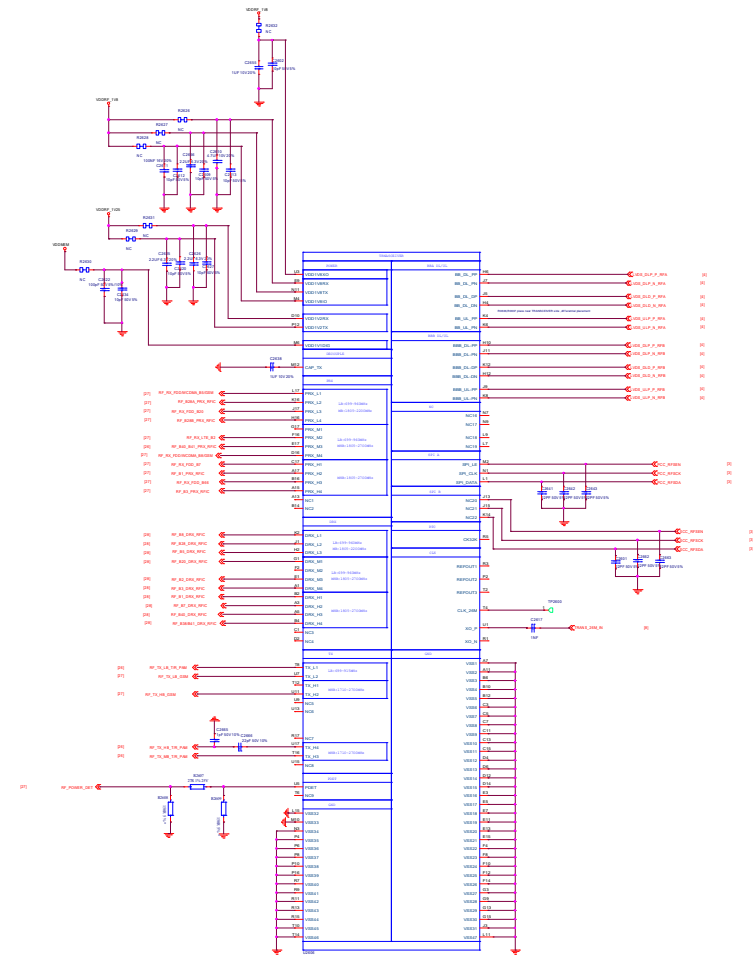
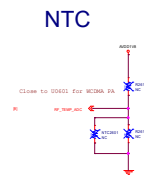
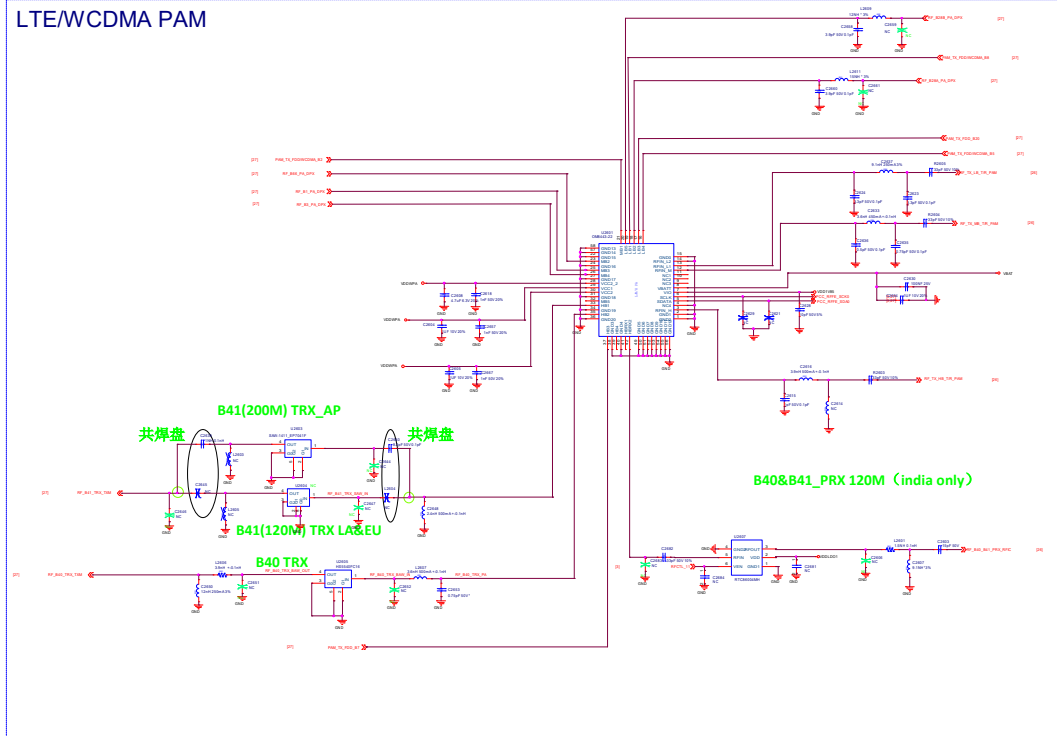
15:+C3053

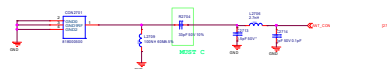
16:删除物料R1801、R1802

17:调换C3109、C3119至NFC芯片端的网络

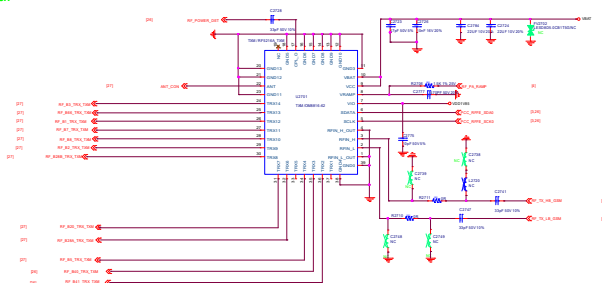
25_RF_change list		Rev: 00.1
Document No.: LONGCHENG-4-1007	Rev: 00	
Department: DEPARTMENT	Department: DEPARTMENT	Division: DIVISION
LONGCHENG		LONGCHENG
Rev: 00.00.00.00		Rev: 00.00.00.00

LTE/WCDMA PAM

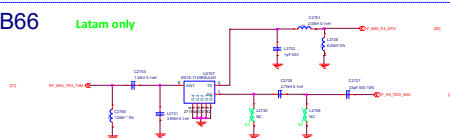




TDD B34/B39/GSM PA
+ Switch

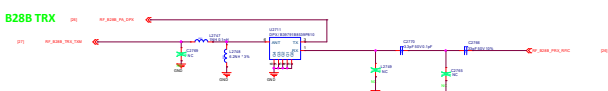


B4/B66 Latam only

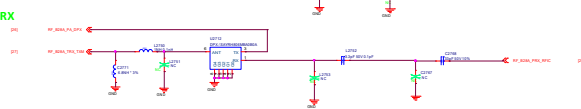


Latam&CIS

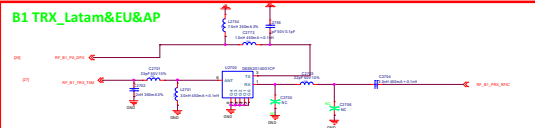
B28B TRX



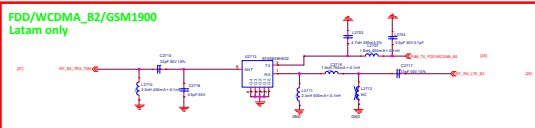
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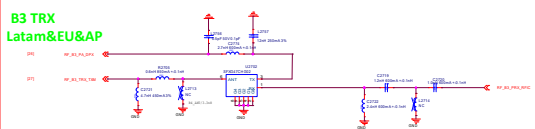
B1 TRX_Latam&EU&AP



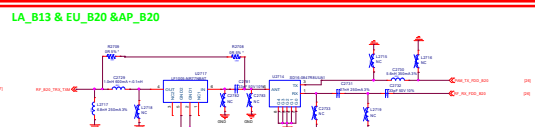
FDD/WCDMA_B2/GSM1900
Latam only



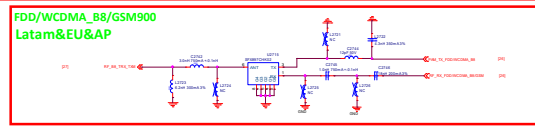
B3 TRX
Latam&EU&AP



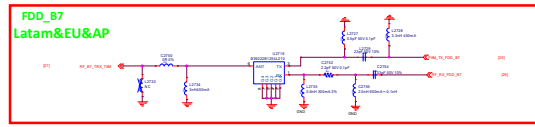
LA_B13 & EU_B20 & AP_B20



FDD/WCDMA_B8/GSM900
Latam&EU&AP



FDD_B7
Latam&EU&AP



FDD/WCDMA_B5/GSM850
Latam_B5_B26&EU_B5&AP_B5_B18_B19_B26

